

The Legacy of PRIME MINISTERS

The ATV project has remained under the direct supervision and control of Indian prime ministers

INDIRA GANDHI 1966-77 and 1980-84

Authorises building of an N-sub (SSBN) after 1974 Pokhran test. Launches ATV programme in March 1984. Signs IGA with the Soviet Union in 1984 for assistance in design and construction of an N-sub.

RAJIV GANDHI 1984-89

Oversees ATV programme in its early years. Understood its relevance to the N-weapon programme he restarted. Became the only Indian PM to visit an N-sub, the INS *Chakra*, leased from Russia in 1988.

ATAL BIHARI VAJPAYEE 1998-2004

Conducts N-tests, enunciates nuclear doctrine which calls for a submarine-based second-strike capability. Reopens talks with Russia for N-sub lease to train crews and assistance on the reactor. Deal for leasing one Akula-2 N-sub signed.

MANMOHAN SINGH 2004 till date

Supervises the final phase of ATV construction. Enlarges the N-sub programme with sanction for building of a new class of SSNs and SSBNs based on Russia's Akula class for a projected fleet of 10 N-subs over the next 20 years.

has 10 nuclear-powered submarines and is building an equal number, giving the Chinese navy tremendous reach into the Indian Ocean. But India has still a long way to go, says strategic affairs expert Brahma Chellaney. "It will still be some years before an N-sub with SLBMs is deployed. In fact, the gaps in India's nuclear deterrent vis-à-vis China remain glaring. If India's nuclear deterrent was credible, Beijing wouldn't mess with India. But the rising Chinese bellicosity suggests otherwise," he says.

The *Arihant* has taken 11 years to complete. It is the first in a series of nuclear-powered submarines to be built over the next two decades. The long arduous road began in 1967 with a Bhabha Atomic Research Centre (BARC) feasibility report on nuclear propulsion. A more detailed report was presented in 1971. And after the Pokhran nuclear test of 1974, Indira Gandhi authorised a project to build a nuclear submarine which would carry a robust, survivable nuclear deterrent. It was always called a naval reactor project. For

a good reason though. Compacting a nuclear reactor to fit snugly within the submarine's 10-metre diameter steel cylinder was going to be the greatest challenge. The reactor also had to go from full ahead to full astern and also from high speed to low speed. The BARC derivatives of its civilian power reactors were too large and incapable of meeting the required performance parameters. Work on the ATV began only in the 1980s with Soviet assistance.

In 1981, Indira Gandhi sent a joint navy and BARC team to visit the USSR to study an offer from the Soviet Union to design and build nuclear submarine. "She was enthused by the fact that we were getting access to so much high-

technology," says a former project head. Months before her assassination in 1984, Indira Gandhi supervised a secret intergovernmental agreement with the Soviet Union under which India would receive training on handling a nuclear submarine and design assistance to build one. A three-year lease was signed for acquiring an elderly Charlie-I class nuclear-powered cruise missile submarine (SSGN). And the ATV project team was set up, headed by a retired vice-admiral, who was given the rank of secretary to the Government of India who and reported to the chief of Defence Research and Development Organisation (DRDO).

The first group from the navy's nas-

ALONG WITH MISSILES AND AIRCRAFT, THE *ARIHANT* COMPLETES THE TRIAD OF DELIVERING NUCLEAR WEAPONS.